作业（共2页，第一页）

- 分布函数:
  1. 教材第三章 6,
  2. 教材第三章 7(b),
  3. 教材第三章 8,
  4. 教材第三章 11.

作业（共2页，第二页）

- 随机变量函数的分布：

5. 设$X \sim \mathcal{U}(-1,2)$，求$Y = X^2$的概率分布。
6. 设X的概率密度(PDF)为

$$f(x) = \frac{1}{\pi(1+x^2)}, -\infty < x < +\infty.$$

① 求X的CDF。

② 求$Y = \arctan(X)$的CDF和PDF，并给出Y的分布名字及对应参数。

7. 假设$X \sim N(0, \sigma^2)$，求$Y = \sqrt{\max(X, 0)}$的概率分布。

思考题

- Marginal => Joint?
  - 存在性
  - 唯一性
- 离散型：
  - 假设我们有两个离散型随机变量X_1,X_2. 是否(X_1,X_2)是一个离散型随机向量，即是否(X_1,X_2)存在联合分布列(Joint PMF)?
  - 假若(X_1,X_2)存在联合分布列(Joint PMF)但未知. 目前已知X_1,X_2各自的边缘分布列(Marginal PMF)，是否可以从这两个边缘分布列得到(X_1,X_2)的联合分布列?
- 连续型：
  - 假设我们有两个连续型随机变量X_1,X_2. 是否(X_1,X_2)是一个连续型随机向量，即是否(X_1,X_2)存在联合密度函数(Joint PDF)?
  - 假若(X_1,X_2)存在联合密度函数(Joint PDF)但未知. 目前已知X_1,X_2各自的边缘密度函数(Marginal PDF)，是否可以从这两个边缘密度函数得到(X_1,X_2)的联合密度函数?

教材题目的具体内容

作业：分布函数

- Chapter3

Problem 6. Calamity Jane goes to the bank to make a withdrawal, and is equally likely to find 0 or 1 customers ahead of her. The service time of the customer ahead, if present, is exponentially distributed with parameter λ. What is the CDF of Jane's waiting time?

Problem 7. Alvin throws darts at a circular target of radius r and is equally likely to hit any point in the target. Let X be the distance of Alvin's hit from the center.

(b) The target has an inner circle of radius t. If $X \leq t$, Alvin gets a score of $S = 1/X$. Otherwise his score is $S = 0$. Find the CDF of S. Is S a continuous random variable?

作业：分布函数

Problem 8. Consider two continuous random variables Y and Z, and a random variable X that is equal to Y with probability p and to Z with probability $1-p$.

(a) Show that the PDF of X is given by

$$f_X(x) = pf_Y(x) + (1-p)f_Z(x).$$

(b) Calculate the CDF of the two-sided exponential random variable that has PDF given by

$$f_X(x) = \begin{cases} p\lambda e^{\lambda x}, & \text{if } x < 0, \\ (1-p)\lambda e^{-\lambda x}. & \text{if } x \geq 0, \end{cases}$$

where $\lambda > 0$ and $0 < p < 1$.

Problem 11. Let X and Y be normal random variables with means 0 and 1, respectively, and variances 1 and 4, respectively.

(a) Find $\mathbf{P}(X \leq 1.5)$ and $\mathbf{P}(X \leq -1)$.

(b) Find the PDF of $(Y-1)/2$.

(c) Find $\mathbf{P}(-1 \leq Y \leq 1)$.